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POLISHING COMPOSITION, POLISHING METHOD, AND METHOD FOR MANUFACTURING SEMICONDUCTOR SUBSTRATE

Abstract

The present disclosure is to provide means capable of improving a polishing removal rate of an object to be polished (particularly, silicon oxide) and reducing scratches on a surface of a polished object to be polished (particularly, silicon oxide). The present disclosure provides a polishing composition containing abrasive grains in which an average primary particle size is 1 nm or more and 150 nm or less, and a dispersing medium, in which an aggregation rate of the abrasive grains is 40% or more and less than 75%.

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Background/Summary

BACKGROUND

1. Technical Field

[0001] The present disclosure relates to a polishing composition, a polishing method, and a method for manufacturing a semiconductor substrate.

2. Description of Related Arts

[0002] In recent years, in accordance with multilayer wiring on a surface of a semiconductor substrate, when manufacturing a device, semiconductor substrates are physically polished and planarized, that is, so-called chemical mechanical polishing (CMP) technology is utilized. CMP is a method of planarizing a surface of an object to be polished such as a semiconductor substrate using a polishing composition (slurry) containing abrasive grains such as silica, alumina, and ceria, an anticorrosive agent, a surfactant, and the like, and the object to be polished is wiring, a plug, or the like made of silicon, polysilicon, silicon oxide (SiO_2), carbon-containing silicon oxide (SiOC), silicon nitride (SiN), metal, or the like.

[0003] For example, JP-A-2016-56292 discloses a polishing composition which contains abrasive grains and at least one of polyacrylic acid and a polyacrylic acid derivative, and in which an electrical conductivity is 2.0 mS/cm or more. According to JP-A-2016-56292, silicon oxide can be polished at a high polishing removal rate.

SUMMARY

[0004] However, the technique described in JP-A-2016-56292 still has room for improvement in terms of reducing scratches on the surface of the polished object to be polished.

[0005] Therefore, an object of the present disclosure is to provide means capable of improving the polishing removal rate of the object to be polished (particularly, silicon oxide) and reducing scratches on the surface of the polished object to be polished (particularly, silicon oxide).

[0006] The inventor of the present disclosure has conducted intensive studies which may solve the above problems. As a result, the present inventor has found that the above problems may be solved by a polishing composition containing abrasive grains in which an average primary particle size is 1 nm or more and 150 nm or less, and a dispersing medium, in which an aggregation rate of the abrasive grains is 40% or more and less than 75%, and have completed the invention of the present disclosure.

Description

DETAILED DESCRIPTION

[0007] According to an embodiment of the present disclosure, there is provided a polishing composition containing abrasive grains in which an average primary particle size is 1 nm or more and 150 nm or less, and a dispersing medium, in which an aggregation rate of the abrasive grains is 40% or more and less than 75%. According to such a polishing composition of the present disclosure, the polishing removal rate of the object to be polished (particularly, silicon oxide) can be improved, and scratches on the surface of the polished object to be polished (particularly, silicon oxide) can be reduced.

[0008] Hereinafter, embodiments of the present disclosure will be described, but the present disclosure is not limited only to the following embodiments, and various modifications can be made within the scope of claims. The embodiments described in the present specification may be combined in any manner to make other embodiments. In the present specification, unless otherwise specified, operations and measurements of physical properties and the like are performed under conditions of room temperature (20° C. or more and 25° C. or less)/relative humidity of 40% RH or

more and 50% RH or less.

[Abrasive Grains]

[0009] The polishing composition according to the present disclosure contains abrasive grains. The abrasive grains have an action of mechanically polishing the object to be polished, and improve the polishing removal rate of the object to be polished by the polishing composition.

[0010] The type of abrasive grain is not particularly limited, and examples thereof include metal oxides such as silica, alumina, zirconia, and titania. The abrasive grains can be used alone or in combination of two or more types thereof. As the abrasive grains, a commercially available product or a synthetic product may be used.

[0011] The type of abrasive grain is preferably silica, and more preferably colloidal silica.

Examples of the method for manufacturing colloidal silica include a sodium silicate method and a sol-gel method, and any colloidal silica manufactured by any manufacturing method is suitably used as the abrasive grains according to the present disclosure. However, from the viewpoint of reducing metal impurities, colloidal silica manufactured by a sol-gel method, which can manufacture colloidal silica with high purity, is preferable.

[0012] The manufacturing of colloidal silica by the sol-gel method can be performed using a conventionally known method, and specifically, colloidal silica can be obtained by performing a hydrolysis/condensation reaction using a hydrolyzable silicon compound (for example, an alkoxysilane or a derivative thereof) as a raw material.

[0013] The shape of the abrasive grains is not particularly limited, and may be spherical or non-spherical. Specific examples of the non-spherical shape include various shapes such as a polygonal columnar shape such as a triangular prism and a quadrangular prism, a columnar shape, a barrel shape in which the central portion of the cylinder bulges more than the end portions, a donut shape in which a central portion of a disk penetrates, a plate shape, a so-called cocoon shape having a constriction at the central portion, a so-called associated spherical shape in which a plurality of particles are integrated, a so-called Konpeito shape having a plurality of protrusions, and a rugby ball shape, and are not particularly limited.

[0014] The average primary particle size of the abrasive grains according to the present disclosure is 1 nm or more and 150 nm or less. When the average primary particle size of the abrasive grains is less than 1 nm, the polishing removal rate of the object to be polished (particularly, silicon oxide) decreases. On the other hand, in a case where the average primary particle size of the abrasive grains exceeds 150 nm, scratches on the surface of the polished object to be polished (particularly, silicon oxide) increase. The average primary particle size of the abrasive grains is preferably 5 nm or more, more preferably 8 nm or more, still more preferably 10 nm or more, and particularly preferably 12 nm or more. As the average primary particle size of the abrasive grains increases, the polishing removal rate of the object to be polished by the polishing composition is improved. In addition, the average primary particle size of the abrasive grains is preferably 100 nm or less, more preferably 80 nm or less, still more preferably 60 nm or less, and particularly preferably 50 nm or less. As the average primary particle size of the abrasive grains decreases, it becomes easier to obtain a surface with fewer defects by polishing using the polishing composition. That is, the average primary particle size of the abrasive grains is preferably 5 nm or more and 100 nm or less, more preferably 8 nm or more and 80 nm or less, still more preferably 10 nm or more and 60 nm or less, and particularly preferably 12 nm or more and 50 nm or less. The average primary particle size of the abrasive grains can be calculated based on, for example, the specific surface area (SA) of the abrasive grains calculated from the BET method on the assumption that the shape of the abrasive grains is a true sphere. In the present specification, as the average primary particle size of the abrasive grains, a value measured by the method described in Examples is adopted.

[0015] In addition, the average secondary particle size of the abrasive grains is preferably 10 nm or more, more preferably 15 nm or more, still more preferably 20 nm or more, and particularly preferably 25 nm or more. As the average secondary particle size of the abrasive grains increases,

resistance during polishing decreases, and polishing can be stably performed. In addition, the average secondary particle size of the abrasive grains is preferably 400 nm or less, more preferably 300 nm or less, still more preferably 200 nm or less, and particularly preferably 100 nm or less. As the average secondary particle size of the abrasive grains decreases, the surface area per unit mass of the abrasive grains increases, the contact frequency with the object to be polished is improved, and the polishing removal rate is further improved. That is, the average secondary particle size of the abrasive grains is preferably 10 nm or more and 400 nm or less, more preferably 15 nm or more and 300 nm or less, still more preferably 20 nm or more and 200 nm or less, and particularly preferably 25 nm or more and 100 nm or less. The average secondary particle size of the abrasive grains can be measured by, for example, a dynamic light scattering method represented by a laser diffraction scattering method.

[0016] That is, the average association degree of the abrasive grains is preferably 5.0 or less, more preferably 4.0 or less, still more preferably 3.0 or less, and particularly preferably 2.5 or less. As the average association degree of the abrasive grains decreases, defects can be further reduced. The average association degree of the abrasive grains is preferably 1.0 or more, more preferably 1.5 or more, and still more preferably 2.0 or more. The average association degree is obtained by dividing the value of the average secondary particle size of the abrasive grains by the value of the average primary particle size. As the average association degree of the abrasive grains increases, there is an advantageous effect that the polishing removal rate of the object to be polished by the polishing composition is improved.

[0017] The upper limit of the aspect ratio of the abrasive grains in the polishing composition is not particularly limited, but is preferably less than 2.0, more preferably 1.8 or less, and still more preferably 1.5 or less. Within such a range, defects on the surface of the object to be polished can be further reduced. The aspect ratio is an average of values obtained by taking the smallest rectangle circumscribing the image of the abrasive grains by the scanning electron microscope and dividing the length of the long side of the rectangle by the length of the short side of the same rectangle, and can be obtained using general image analysis software. The lower limit of the aspect ratio of the abrasive grains in the polishing composition is not particularly limited, but is preferably 1.0 or more, and more preferably 1.2 or more.

[0018] In the particle size distribution of the abrasive grains determined by a laser diffraction scattering method, the lower limit of D90/D10, which is the ratio between the particle size (D90) when the integrated particle mass reaches 90% of the total particle mass from the fine particle side and the particle size (D10) when the total particle mass reaches 10% of the total particle mass, is not particularly limited, but is preferably 1.1 or more, more preferably 1.4 or more, still more preferably 1.7 or more, and most preferably 2.0 or more. In addition, in the particle size distribution of the abrasive grains in the polishing composition determined by a laser diffraction scattering method, the upper limit of the ratio D90/D10 of the particle size (D90) when the integrated particle mass reaches 90% of the total particle mass from the fine particle side to the particle size (D10) when the total particle mass reaches 10% of the total particle mass is not particularly limited, but is preferably 3.0 or less, and more preferably 2.5 or less. Within such a range, defects on the surface of the object to be polished can be further reduced.

[0019] The size (average primary particle size, average secondary particle size, aspect ratio, D90/D10, and the like) of the abrasive grains can be appropriately controlled by selecting a method for manufacturing the abrasive grains, for example.

<Aggregation Rate of Abrasive Grains>

[0020] In the present disclosure, the aggregation rate of the abrasive grains is 40% or more and less than 75%. By using the abrasive grains having the aggregation rate in such a range, the polishing removal rate of the object to be polished (particularly, silicon oxide) can be improved, and scratches on the surface of the polished object to be polished (particularly, silicon oxide) can be reduced. In a case where the aggregation rate is less than 40%, the polishing removal rate of the

object to be polished (particularly, silicon oxide) decreases. On the other hand, in a case where the aggregation rate is 75% or more, scratches on the surface of the polished object to be polished (particularly, silicon oxide) increase. The aggregation rate is preferably 40% or more and 55% or less.

[0021] In the present specification, as the aggregation rate of the abrasive grains, a value measured by the following method is adopted:

<<Method for Measuring Aggregation Rate>>

[0022] 1) 30.0 g of the polishing composition as a sample was weighed into a polypropylene container, and the upper lid was firmly closed; [0023] 2) Using a centrifuge, the operation was performed continuously for 15 minutes at a temperature of 25° C. and 15,000 rpm; [0024] 3) By centrifugation, the moisture at the top of the container and the abrasive grains at the bottom were separated, and the abrasive grains were compressed and settled at the bottom of the container (referred to as a “cake”); [0025] 4) The settled cake was dried at 200° C. for 24 hours using a dryer, then the weight of the obtained solid was measured; and [0026] 5) Separately, using an aqueous dispersion containing only abrasive grains, the weight of the dried solid measured by the procedures of 1) to 4) above was defined as (B), the weight of the dried solid measured using the polishing composition as a sample was defined as (A), and the aggregation rate was calculated by the following Formula (1).

[MathematicalFormula1]

[00001]
$$\text{Aggregationrate(\%)} = [(A) / (B)] \times 100 \quad (1)$$

[0027] (A) and (B) may contain a trace amount of additive, but a value calculated by Formula (1) without considering this is taken as the aggregation rate of the abrasive grains. More specifically, measurement is performed by the method described in Examples.

[0028] The aggregation rate of the abrasive grains can be controlled by selecting the type and addition amount of the inorganic salt and the organic onium salt to be described later, performing filtration of the polishing composition, adjusting the abrasive grain concentration, and the like.

[0029] In the polishing composition according to the present disclosure, the abrasive grains preferably have a negative zeta potential. Here, the “zeta (C) potential” is a potential difference generated at an interface between a solid and a liquid in contact with each other when the solid and the liquid perform relative movement.

[0030] The zeta potential of the abrasive grains in the polishing composition according to the present disclosure is preferably −60 mV or more and −10 mV or less, more preferably −50 mV or more and −10 mV or less, still more preferably −40 mV or more and −15 mV or less, and particularly preferably more than −35 mV and −15 mV or less. Since the abrasive grains have a zeta potential in such a range, the polishing removal rate of the object to be polished can be further improved. Here, the zeta potential of the abrasive grains in the polishing composition is a value measured by the method described in Examples. In addition, the zeta potential of the abrasive grains can be adjusted by the amount of an anionic group (particularly, an organic acid group) of the abrasive grains described below, the pH of the polishing composition, and the like.

[0031] In some embodiments of the present disclosure, the colloidal silica contained in the polishing composition is preferably anionically modified colloidal silica (anion-modified colloidal silica), and more preferably colloidal silica with an organic acid immobilized on the surface. Colloidal silica with an organic acid immobilized on the surface tends to have a larger absolute value of zeta potential in the polishing composition than normal colloidal silica with no organic acid immobilized thereon. Therefore, it is easy to adjust the zeta potential of the colloidal silica in the polishing composition to negative (for example, in the range of −40 mV or more and −15 mV or less).

[0032] Preferable examples of the colloidal silica with an organic acid immobilized on the surface include colloidal silica with an organic acid group such as a carboxylic acid group, a sulfonic acid

group, a phosphonic acid group, or an aluminate group immobilized on the surface. Among them, from the viewpoint of easy manufacturing, colloidal silica with sulfonic acid and carboxylic acid immobilized on the surface is preferable, and colloidal silica with sulfonic acid immobilized on the surface is more preferable.

[0033] The immobilization of the organic acid on the surface of the colloidal silica is not achieved by simply allowing the colloidal silica and the organic acid to be present together. For example, when a sulfonic acid, which is a type of organic acid, is immobilized on colloidal silica, the immobilization can be performed by, for example, the method described in “Sulfonic acid-functionalized silica through quantitative oxidation of thiol groups”, Chem. Commun. 246-247 (2003). Specifically, a silane coupling agent having a thiol group such as 3-mercaptopropyltrimethoxysilane is coupled to colloidal silica, and then the thiol group is oxidized with hydrogen peroxide, whereby colloidal silica (sulfonic acid-modified colloidal silica) with sulfonic acid immobilized on the surface can be obtained.

[0034] Alternatively, when a carboxylic acid, which is a type of organic acid, is immobilized on colloidal silica, the immobilization can be performed, for example, by the method described in “Novel Silane Coupling Agents Containing a Photolabile 2-Nitrobenzyl Ester for Introduction of a Carboxy Group on the Surface of Silica Gel”, Chemistry Letters, 3, 228-229 (2000). Specifically, a silane coupling agent containing a photoreactive 2-nitrobenzyl ester is coupled to colloidal silica and then irradiated with light, whereby colloidal silica (carboxylic acid-modified colloidal silica) with a carboxylic acid immobilized on the surface can be obtained.

[0035] The concentration (content) of the abrasive grains is not particularly limited, but is preferably 0.5% by mass or more, more preferably 0.8% by mass or more, still more preferably 1% by mass or more, further preferably more than 1% by mass, and particularly preferably 1.5% by mass or more, with respect to the total mass of the polishing composition. In addition, the upper limit of the concentration (content) of the abrasive grains is preferably 20% by mass or less, more preferably 15% by mass or less, still more preferably 10% by mass or less, and particularly preferably 5% by mass or less, with respect to the total mass of the polishing composition. That is, the concentration (content) of the abrasive grains is preferably 0.5% by mass or more and 20% by mass or less, more preferably 0.8% by mass or more and 20% by mass or less, still more preferably 1% by mass or more and 15% by mass or less, further more preferably more than 1% by mass and 10% by mass or less, and particularly preferably 1.5% by mass or more and 5% by mass or less, with respect to the total mass of the polishing composition. Within such a range, the polishing removal rate can be improved while suppressing the cost. In addition, in a case where the polishing composition contains two or more types of abrasive grains, the concentration (content) of the abrasive grains means the total amount thereof.

[Inorganic Salt]

[0036] The polishing composition according to the present disclosure preferably contains an inorganic salt. The inorganic salt has a function of reducing scratches on the surface of the polished object to be polished (particularly, silicon oxide). In addition, the inorganic salt has a function of increasing the electrical conductivity of the polishing composition to further improve the polishing removal rate of the object to be polished (particularly, silicon oxide). Furthermore, the inorganic salt has a function of controlling the aggregation rate of the abrasive grains.

[0037] Examples of the inorganic salt include inorganic salts composed of a cation and an anion shown below. Examples of the cation include alkali metal ions such as lithium ions, sodium ions, and potassium ions, alkaline earth metal ions such as magnesium ions, calcium ions, and strontium ions, polyatomic ions such as ammonium ions, complex ions, and the like. Examples of the anion include a halide ion (fluoride ion, chloride ion, bromide ion, iodide ion, and the like), an oxoacid ion (a borate ion, a carbonate ion, a nitrate ion, a nitrite ion, a metasilicate ion, a phosphate ion, a monohydrogen phosphate ion, a dihydrogen phosphate ion, a phosphonate ion, a monohydrogen phosphonate ion, a phosphinate ion, a sulfate ion, a sulfonate ion, a sulfite ion, a thiosulfate ion, a

chromate ion, a dichromate ion, a permanganate ion, and the like), a thiocyanate ion, a cyanate ion, a sulfamate ion, and the like.

[0038] More specific examples of the inorganic salt include lithium salts such as lithium chloride, lithium bromide, lithium carbonate, lithium nitrate, and lithium thiocyanate; calcium salts such as calcium chloride, calcium bromide, calcium carbonate, calcium nitrate and calcium thiocyanate; iron salts such as iron nitrate and iron thiocyanate; potassium salts such as potassium chloride, potassium bromide, potassium nitrate, potassium sulfate, potassium thiocyanate, potassium sulfamate, potassium phosphate, potassium dihydrogen phosphate, potassium monohydrogen phosphate, and potassium monohydrogen phosphonate; sodium salts such as sodium chloride, sodium bromide, sodium nitrate, sodium sulfate, and sodium thiocyanate; zinc salts such as zinc chloride, zinc nitrate, and zinc thiocyanate; magnesium salts such as magnesium nitrate, magnesium sulfate, and magnesium thiocyanate; strontium salts such as strontium nitrate and strontium thiocyanate; and ammonium salts such as ammonium chloride, ammonium bromide, ammonium iodide, ammonium nitrate, ammonium phosphate, ammonium dihydrogen phosphate, ammonium monohydrogen phosphate, ammonium phosphonate, ammonium monohydrogen phosphonate, ammonium sulfate, ammonium thiocyanate, ammonium sulfamate, and the like.

These inorganic salts may be used alone or in combination of two or more types thereof. As the inorganic salt, a commercially available product may be used, or a synthetic product may be used.

[0039] Among them, the inorganic salt is preferably at least one of an ammonium salt of an inorganic acid or a potassium salt of an inorganic acid from the viewpoint of further exhibiting the effect of the present disclosure. The inorganic acid is preferably sulfuric acid, nitric acid, or carbonic acid. Therefore, the inorganic salt is more preferably at least one selected from the group consisting of ammonium sulfate, ammonium nitrate, ammonium carbonate, potassium sulfate, potassium nitrate, and potassium carbonate, and still more preferably ammonium sulfate.

[0040] The concentration (content) of the inorganic salt in the polishing composition is not particularly limited, but in the case of a polishing composition used for polishing an object to be polished as it is as a polishing liquid, the lower limit of the concentration (content) of the inorganic salt in the polishing composition is preferably 0.005% by mass (50 ppm by mass) or more, more preferably 0.01% by mass (100 ppm by mass) or more, still more preferably 0.1% by mass (1,000 ppm by mass) or more, and particularly preferably 0.5% by mass (5,000 ppm by mass) or more, with respect to the total mass of the polishing composition. In addition, the upper limit of the concentration (content) of the inorganic salt in the polishing composition is preferably 2.0% by mass (20,000 ppm by mass) or less, more preferably 1.5% by mass (15,000 ppm by mass) or less, still more preferably 1.3% by mass (13,000 ppm by mass) or less, and particularly preferably 1.0% by mass (10,000 ppm by mass) or less with respect to the total mass of the polishing composition.

[0041] That is, the concentration (content) of the inorganic salt is preferably 0.005% by mass (50 ppm by mass) or more and 2.0% by mass (20,000 ppm by mass) or less, more preferably 0.01% by mass (100 ppm by mass) or more and 1.5% by mass (15,000 ppm by mass) or less, still more preferably 0.1% by mass (1,000 ppm by mass) or more and 1.3% by mass (13,000 ppm by mass) or less, and particularly preferably 0.5% by mass (5,000 ppm by mass) or more and 1.0% by mass (10,000 ppm by mass) or less, with respect to the total mass of the polishing composition.

[0042] When the polishing composition contains two or more types of inorganic salts, the concentration (content) of the inorganic salts means the total amount thereof.

[Organic Onium Salt]

[0043] The polishing composition according to the present disclosure preferably contains an organic onium salt. The organic onium salt has a function of reducing scratches on the surface of the polished object to be polished (particularly, silicon oxide). In addition, the organic onium salt has a function of controlling the aggregation rate of the abrasive grains.

[0044] The organic onium salt used in the present disclosure preferably is at least one of a tetraalkylammonium salt represented by the following Chemical Formula 1 or a

tetraalkylphosphonium salt represented by the following Chemical Formula 2.

[Chem. 1]

[NR.¹R.²R.³R.⁴].⁺A.⁻ Chemical Formula 1

[PR.⁵R.⁶R.⁷R.⁸].⁺X.⁻ Chemical Formula 2

[0045] In the Chemical Formula 1 and the Chemical Formula 2, [0046] R.¹ to R.⁸ each independently represent an unsubstituted alkyl group having 1 or more and 4 or less carbon atoms, and [0047] A.⁻ and X.⁻ each independently represent a monovalent anion.

[0048] When an organic onium salt having an alkyl group having 5 or more carbon atoms is used, scratches on the surface of the polished object to be polished may increase.

[0049] Specific examples of the unsubstituted alkyl group having 1 or more and 4 or less carbon atoms used for R.¹ to R.⁸ in the above Chemical Formulas 1 and 2 include a methyl group, an ethyl group, a n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, and a tert-butyl group. From the viewpoint that the effect of the present disclosure is further exhibited, an unsubstituted alkyl group having 2 or more and 4 or less carbon atoms, such as an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, and a tert-butyl group, is preferable.

[0050] Examples of the monovalent anion used for A.⁻ and X.⁻ in the above Chemical Formulas 1 and 2 are not particularly limited, but are halide ions such as a fluoride ion, a chloride ion, a bromide ion, and an iodide ion; hydroxide ion; organic acid ions such as benzoate ions; and the like are suitable. The monovalent anion may be used alone or in combination of two or more types thereof. From the viewpoint that the effect of the present disclosure is more exhibited, A.⁻ and X.⁻ in the above Chemical Formulas 1 and 2 are preferably hydroxide ions (OH.⁻).

[0051] More specific examples of the tetraalkylammonium salt represented by the above Chemical Formula 1 include tetramethylammonium fluoride, trimethylethylammonium fluoride, dimethyldiethylammonium fluoride, methyltriethylammonium fluoride, tetraethylammonium fluoride, trimethyl n-propylammonium fluoride, trimethylisopropylammonium fluoride, dimethylethyl n-propylammonium fluoride, dimethylethyl isopropylammonium fluoride, methyl diethyl n-propylammonium fluoride, methyl diethylisopropylammonium fluoride, triethyl isopropylammonium fluoride, triethyl n-propylammonium fluoride, tetra n-propylammonium fluoride, tetraisopropylammonium fluoride, tetra n-butylammonium fluoride, and tetra tert-butylammonium fluoride; tetramethylammonium chloride, trimethylethylammonium chloride, dimethyldiethylammonium chloride, methyltriethylammonium chloride, tetraethylammonium chloride, trimethyl n-propylammonium chloride, trimethylisopropylammonium chloride, dimethylethyl n-propylammonium chloride, dimethylethylisopropylammonium chloride, methyldiethyl n-propylammonium chloride, methyldiethylisopropylammonium chloride, triethylisopropylammonium chloride, triethyl n-propylammonium chloride, tetra n-propylammonium chloride, tetraisopropylammonium chloride, tetra n-butylammonium chloride, and tetra tert-butylammonium chloride; tetramethylammonium bromide, trimethylethylammonium bromide, dimethyldiethylammonium bromide, methyltriethylammonium bromide, tetraethylammonium bromide, trimethyl n-propylammonium bromide, trimethylisopropylammonium bromide, dimethylethyl n-propylammonium bromide, dimethylethylisopropylammonium bromide, methyldiethyl n-propylammonium bromide, methyldiethylisopropylammonium bromide, triethylisopropylammonium bromide triethyl n-propylammonium bromide, tetra n-propylammonium bromide, tetraisopropylammonium bromide, tetra n-butylammonium bromide, and tetra tert-butylammonium bromide; tetramethylammonium iodide, trimethylethylammonium iodide, dimethyldiethylammonium iodide, methyltriethylammonium iodide, tetraethylammonium iodide, trimethyl n-propylammonium

iodide, trimethylisopropylammonium iodide, dimethylethyl n-propylammonium iodide, dimethylethylisopropylammonium iodide, methyldiethyl n-propylammonium iodide, methyldiethylisopropylammonium iodide, triethylisopropylammonium iodide, triethyl n-propylammonium iodide, tetra n-propylammonium iodide, tetraisopropylammonium iodide, tetra n-butylammonium iodide, and tetra tert-butylammonium iodide; tetramethylammonium hydroxide, trimethylethylammonium hydroxide, dimethyldiethylammonium hydroxide, methyltriethylammonium hydroxide, tetraethylammonium hydroxide, trimethyl n-propylammonium hydroxide, trimethylisopropylammonium hydroxide, dimethylethyl n-propylammonium hydroxide, dimethylethylisopropylammonium hydroxide, methyldiethyl n-propylammonium hydroxide, methyldiethylisopropylammonium hydroxide, triethylisopropylammonium hydroxide, triethyl n-propylammonium hydroxide, tetra n-propylammonium hydroxide, tetraisopropylammonium hydroxide, tetra n-butylammonium hydroxide, and tetra tert-butylammonium hydroxide; and tetramethylammonium benzoate, trimethylethylammonium benzoate, dimethyldiethylammonium benzoate, methyltriethylammonium benzoate, tetraethylammonium benzoate, trimethyl n-propylammonium benzoate, trimethylisopropylammonium benzoate, dimethylethyl n-propylammonium benzoate, dimethylethylisopropylammonium benzoate, methyldiethyl n-propylammonium benzoate, methyldiethylisopropylammonium benzoate, triethylisopropylammonium benzoate, triethyl n-propylammonium benzoate, tetra n-propylammonium benzoate, tetraisopropylammonium benzoate, tetra n-butylammonium benzoate, tetra tert-butylammonium benzoate, and the like.

[0052] More specific examples of the tetraalkylphosphonium salt represented by the above Chemical Formula 2 include tetramethylphosphonium fluoride, trimethylethylphosphonium fluoride, dimethyldiethylphosphonium fluoride, methyltriethylphosphonium fluoride, tetraethylphosphonium fluoride, trimethyl n-propylphosphonium fluoride, trimethylisopropylphosphonium fluoride, dimethylethyl n-propylphosphonium fluoride, dimethylethylisopropylphosphonium fluoride, methyldiethyl n-propylphosphonium fluoride, methyldiethylisopropylphosphonium fluoride, triethylisopropylphosphonium fluoride, triethyl n-propylphosphonium fluoride, tetra n-propylphosphonium fluoride, tetraisopropylphosphonium fluoride, tetra n-butylphosphonium fluoride, and tetra tert-butylphosphonium fluoride; tetramethylphosphonium chloride, trimethylethylphosphonium chloride, dimethyldiethylphosphonium chloride, methyltriethylphosphonium chloride, tetraethylphosphonium chloride, trimethyl n-propylphosphonium chloride, trimethylisopropylphosphonium chloride, dimethylethyl n-propylphosphonium chloride, dimethylethylisopropylphosphonium chloride, methyldiethyl n-propylphosphonium chloride, methyldiethylisopropylphosphonium chloride, triethylisopropylphosphonium chloride, triethyl n-propylphosphonium chloride, tetra n-propylphosphonium chloride, tetraisopropylphosphonium chloride, tetra n-butylphosphonium chloride, tetra tert-butylphosphonium chloride; tetramethylphosphonium bromide, trimethylethylphosphonium bromide, dimethyldiethylphosphonium bromide, methyltriethylphosphonium bromide, tetraethylphosphonium bromide, trimethyl n-propylphosphonium bromide, trimethylisopropylphosphonium bromide, dimethylethyl n-propylphosphonium bromide, dimethylethylisopropylphosphonium bromide, methyldiethyl n-propylphosphonium bromide, methyldiethylisopropylphosphonium bromide, triethylisopropylphosphonium bromide, triethyl n-propylphosphonium bromide, tetra n-propylphosphonium bromide, tetraisopropylphosphonium bromide, tetra n-butylphosphonium bromide, and tetra tert-butylphosphonium bromide; tetramethylphosphonium iodide, trimethylethylphosphonium iodide, dimethyldiethylphosphonium iodide, methyltriethylphosphonium iodide, tetraethylphosphonium iodide, trimethyl n-propylphosphonium iodide, trimethylisopropylphosphonium iodide, dimethylethyl n-propylphosphonium iodide, dimethylethylisopropylphosphonium iodide, methyldiethyl n-propylphosphonium iodide, methyldiethylisopropylphosphonium iodide,

triethylisopropylphosphonium iodide, triethyl n-propylphosphonium iodide, tetra n-propylphosphonium iodide, tetraisopropylphosphonium iodide, tetra n-butylphosphonium iodide, and tetra tert-butylphosphonium iodide; tetramethylphosphonium hydroxide, trimethylethylphosphonium hydroxide, dimethyldiethylphosphonium hydroxide, methyltriethylphosphonium hydroxide, tetraethylphosphonium hydroxide, trimethyl n-propylphosphonium hydroxide, trimethylisopropylphosphonium hydroxide, dimethylethyl n-propylphosphonium hydroxide, dimethylethyl isopropylphosphonium hydroxide, methyldiethyl n-propylphosphonium hydroxide, methyldiethylisopropylphosphonium hydroxide, triethyl isopropylphosphonium hydroxide, triethyl isopropylphosphonium hydroxide, triethyl n-propylphosphonium hydroxide, tetra n-propylphosphonium hydroxide, tetraisopropylphosphonium hydroxide, tetra n-butylphosphonium hydroxide, and tetra tert-butylphosphonium hydroxide; and tetramethyl phosphonium benzoate, trimethylethyl phosphonium benzoate, dimethyl diethyl phosphonium benzoate, methyl triethyl phosphonium benzoate, tetraethyl phosphonium benzoate, trimethyl n-propyl phosphonium benzoate, trimethylisopropyl phosphonium benzoate, dimethylethyl n-propyl phosphonium benzoate, dimethylethyl isopropyl phosphonium benzoate, methyl diethyl n-propyl phosphonium benzoate, methyl diethyl isopropyl phosphonium benzoate, triethyl isopropyl phosphonium benzoate, triethyl n-propyl phosphonium benzoate, tetra n-propyl phosphonium benzoate, tetraisopropyl phosphonium benzoate, tetra n-butyl phosphonium benzoate, and tetra tert-butyl ammonium benzoate.

[0053] These organic onium salts may be used alone or in combination of two or more types thereof. In addition, as the organic onium salt, a commercially available product or a synthetic product may be used.

[0054] Among these organic onium salts, a tetraalkylammonium salt represented by the above Chemical Formula 1 is preferable, and at least one selected from the group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetra n-butylammonium hydroxide is more preferable from the viewpoint of more easily exhibiting the effect of the present disclosure.

[0055] The lower limit of the concentration (content) of the organic onium salt in the polishing composition is preferably 0.001% by mass (10 ppm by mass) or more, more preferably 0.005% by mass (50 ppm by mass) or more, still more preferably 0.007% by mass (70 ppm by mass) or more, and particularly preferably 0.01% by mass (100 ppm by mass) or more. In addition, the upper limit of the concentration (content) of the organic onium salt in the polishing composition is preferably 0.3% by mass (3,000 ppm by mass) or less, more preferably 0.1% by mass (1,000 ppm by mass) or less, still more preferably 0.07% by mass (700 ppm by mass) or less, and particularly preferably 0.05% by mass (500 ppm by mass) or less. That is, the concentration (content) of the organic onium salt in the polishing composition is preferably 0.001% by mass (10 ppm by mass) or more and 0.3% by mass (3,000 ppm by mass) or less, more preferably 0.005% by mass (50 ppm by mass) or more and 0.1% by mass (1,000 ppm by mass) or less, still more preferably 0.007% by mass (70 ppm by mass) or more and 0.07% by mass (700 ppm by mass) or less, and particularly preferably 0.01% by mass (100 ppm by mass) or more and 0.05% by mass (500 ppm by mass) or less.

[0056] When the polishing composition contains two or more types of organic onium salts, the concentration (content) of the organic onium salt means the total amount thereof.

[pH and pH Adjusting Agent]

[0057] The pH of the polishing composition according to the present disclosure is not particularly limited, but is preferably 1.0 or more, more preferably 1.5 or more, and still more preferably 2.0 or more. In addition, the pH is preferably 9.0 or less, more preferably 7.0 or less, still more preferably less than 7.0, further preferably 5.0 or less, and particularly preferably 3.5 or less. That is, the pH of the polishing composition according to the present disclosure is preferably 1.0 or more and 9.0 or less, more preferably 1.5 or more and 7.0 or less, still more preferably 1.5 or more and less than

7.0, further preferably 1.5 or more and 5.0 or less, and particularly preferably 2.0 or more and 3.5 or less.

[0058] The inorganic salt and the organic onium salt contained in the polishing composition according to the present disclosure may have a role as a pH adjusting agent for adjusting the pH of the polishing composition, but may further contain a separate pH adjusting agent depending on the target pH. Such a pH adjusting agent may be either an acid or a base, or may be either an inorganic compound or an organic compound. The pH adjusting agent can be used alone or in combination of two or more types thereof.

[0059] Specific examples of the acid that can be used as the pH adjusting agent include inorganic acids such as hydrochloric acid, sulfuric acid, nitric acid, hydrofluoric acid, boric acid, carbonic acid, hypophosphorous acid, phosphorous acid, and phosphoric acid; organic acids such as formic acid, acetic acid, propionic acid, butyric acid, valeric acid, 2-methylbutyric acid, n-hexanoic acid, 3,3-dimethylbutyric acid, 2-ethylbutyric acid, 4-methylpentanoic acid, n-heptanoic acid, 2-methylhexanoic acid, n-octanoic acid, 2-ethylhexanoic acid, benzoic acid, glycolic acid, salicylic acid, glyceric acid, oxalic acid, malonic acid, succinic acid, glutaric acid, adipic acid, pimelic acid, maleic acid, phthalic acid, malic acid, tartaric acid, citric acid, lactic acid, diglycolic acid, 2-furancarboxylic acid, 2,5-furandicarboxylic acid, 3-furancarboxylic acid, 2-tetrahydrofurancarboxylic acid, methoxyacetic acid, methoxyphenylacetic acid, and phenoxyacetic acid.

[0060] Examples of the base that can be used as the pH adjusting agent include amines such as aliphatic amines and aromatic amines, hydroxides of alkali metals such as sodium hydroxide and potassium hydroxide, hydroxides of Group 2 elements, ammonia, and the like.

[0061] The addition amount of the pH adjusting agent is not particularly limited, and may be appropriately adjusted such that the polishing composition has the desired pH. In addition, the pH of the polishing composition can be measured by, for example, a pH meter, and specifically, can be measured by the method described in Examples.

[Dispersing Medium]

[0062] The polishing composition according to the present disclosure contains a dispersing medium. As the dispersing medium, water; alcohols such as methanol, ethanol, and ethylene glycol; ketones or the like such as acetone; mixtures thereof; and the like can be exemplified. Among these, water is preferable as the dispersing medium. That is, according to a preferred aspect of the present disclosure, the dispersing medium contains water. According to a more preferred aspect of the present disclosure, the dispersing medium is substantially composed of water. Note that the above “substantially” is intended to mean that a dispersing medium other than water can be contained as long as the object of the present disclosure can be achieved, and more specifically, the dispersing medium is preferably composed of 90% by mass or more and 100% by mass or less of water and 0% by mass or more and 10% by mass or less of a dispersing medium other than water, and more preferably composed of 99% by mass or more and 100% by mass or less of water and 0% by mass or more and 1% by mass or less of a dispersing medium other than water. Most preferably, the dispersing medium is water.

[0063] From the viewpoint of suppressing the action of the components contained in the polishing composition from being inhibited, water that does not contain impurities as much as possible is preferable as the dispersing medium, and specifically, pure water, ultrapure water from which impurity ions are removed with an ion exchange resin and then foreign substances are removed through a filter, or distilled water is more preferable.

[Electrical Conductivity of Polishing Composition]

[0064] The electrical conductivity (EC) of the polishing composition according to the present disclosure is not particularly limited, but is preferably 1 mS/cm or more, and more preferably 2 mS/cm or more. In addition, the electrical conductivity (EC) of the polishing composition according to the present disclosure is preferably 25 mS/cm or less, and more preferably 20 mS/cm

or less. That is, the electrical conductivity (EC) of the polishing composition according to the present disclosure is preferably 1 mS/cm or more and 25 mS/cm or less, and more preferably 2 mS/cm or more and 20 mS/cm or less. When the electrical conductivity (EC) of the polishing composition is in such a range, repulsion between the abrasive grains can be appropriately adjusted, and stability can be secured. The electrical conductivity of the polishing composition can be adjusted by the type, amount and the like of a pH adjusting agent or the like. In addition, the electrical conductivity can be measured by the method described in Examples.

[Other Components]

[0065] The polishing composition according to the present disclosure may further contain other components such as a water-soluble polymer, a complexing agent, a metal anticorrosive, an antiseptic agent, an antifungal agent, a reducing agent, and a surfactant, as necessary. Hereinafter, an antiseptic agent and an antifungal agent which are preferable components will be described. The oxidizing agent will also be described.

(Antiseptic Agent and Antifungal Agent)

[0066] Examples of the antiseptic agent and the antifungal agent that can be added to the polishing composition according to the present disclosure include isothiazoline-based antiseptic agents such as 2-methyl-4-isothiazoline-3-one and 5-chloro-2-methyl-4-isothiazoline-3-one, paraoxybenzoic acid esters, phenoxyethanol, and the like. These antiseptic agents and the antifungal agents may be used alone, or may be used in combination of two or more thereof.

[Oxidizing Agent]

[0067] The polishing composition according to the present disclosure preferably does not substantially contain an oxidizing agent. When the polishing composition contains an oxidizing agent, the surface of the object to be polished is oxidized to generate an oxide film, and there is a concern that the polishing time becomes long. Specific examples of the oxidizing agent herein include hydrogen peroxide (H.sub.2O.sub.2), sodium persulfate, ammonium persulfate, sodium dichloroisocyanurate, and the like. The fact that the polishing composition does not substantially contain an oxidizing agent means that the polishing composition does not at least intentionally contain an oxidizing agent. Therefore, a polishing composition inevitably containing a trace amount of an oxidizing agent derived from a raw material, a production method, or the like is included in the concept of the polishing composition that does not substantially contain an oxidizing agent. For example, the concentration (content) of the oxidizing agent in the polishing composition is preferably 0.01% by mass (100 ppm by mass) or less, more preferably less than 0.01% by mass (100 ppm by mass), and still more preferably 0.005% by mass (50 ppm by mass) or less. The lower limit of the concentration (content) of the oxidizing agent is preferably 0% by mass or more, and more preferably 0.0005% by mass (5 ppm by mass) or more.

[Form of Polishing Composition]

[0068] The polishing composition according to the present disclosure is typically supplied to an object to be polished in the form of a polishing liquid containing the polishing composition, and is used for polishing the object to be polished. The polishing composition according to the present disclosure may be, for example, diluted (typically diluted with water) and used as a polishing liquid, or may be used as it is as a polishing liquid. That is, the concept of the polishing composition according to the present disclosure includes both a polishing composition (working slurry) supplied to an object to be polished and used for polishing the object to be polished and a concentrated solution (stock solution of working slurry) diluted and used for polishing. The concentration ratio of the concentrated solution can be, for example, about 2 times or more and 100 times or less on a volume basis, and usually about 3 times or more and 50 times or less is appropriate.

[Object to be Polished]

[0069] The object to be polished according to the present disclosure is not particularly limited, and examples thereof include single crystal silicon, polycrystalline silicon (polysilicon), polycrystalline

silicon doped with an n-type or p-type impurity, amorphous silicon (amorphous silicon), amorphous silicon doped with an n-type or p-type impurity, silicon oxide, silicon nitride, silicon carbonitride (SiCN), a metal, SiGe, a carbon-containing material, and the like.

[0070] Examples of the object to be polished containing silicon oxide include a TEOS type silicon oxide film (hereinafter also simply referred to as “TEOS film”) generated using tetraethyl orthosilicate as a precursor, a high density plasma (HDP) film, an undoped silicate glass (USG) film, a phosphorus silicate glass (PSG) film, a boron-phospho silicate glass (BPSG) film, a rapid thermal oxidation (RTO) film, and the like.

[0071] Examples of the metal include tungsten, copper, aluminum, cobalt, hafnium, nickel, gold, silver, platinum, palladium, rhodium, ruthenium, iridium, osmium, and the like.

[0072] Examples of the carbon-containing material include amorphous carbon, spin-on carbon (SOC), diamond-like carbon (DLC), nanocrystalline diamond, graphene, and the like.

[0073] The object to be polished may be a commercially available product or may be manufactured by a known method.

[0074] Among them, an object to be polished containing silicon oxide is preferable. Therefore, according to a preferred embodiment of the present disclosure, the polishing composition is used for polishing an object to be polished containing silicon oxide.

[Method for Manufacturing Polishing Composition]

[0075] The method for manufacturing the polishing composition according to the present embodiment is not particularly limited, and for example, the polishing composition can be obtained by stirring and mixing abrasive grains, an inorganic salt, an organic onium salt, and other additives added as necessary. Details of each component are as described above.

[0076] The temperature at which each component is mixed is not particularly limited, but is preferably 10° C. or more and 40° C. or less, and heating may be performed in order to increase the rate of dissolution. The mixing time is also not particularly limited as long as the mixture can be uniformly mixed.

[Polishing Method and Method for Manufacturing Semiconductor Substrate]

[0077] As described above, the polishing composition according to the present embodiment is particularly suitably used for polishing an object to be polished containing silicon oxide. Therefore, the present disclosure provides a polishing method for polishing an object to be polished containing silicon oxide with the polishing composition according to the present embodiment. In addition, the present disclosure also provides a method for manufacturing a semiconductor substrate, the method including polishing a semiconductor substrate containing silicon oxide by the polishing method according to the polishing method.

[0078] As the polishing apparatus, it is possible to use a general polishing apparatus in which a holder for holding a substrate or the like having an object to be polished, a motor capable of changing the rotation speed, or the like are attached, and which has a polishing table to which a polishing pad (polishing cloth) can be attached.

[0079] As the polishing pad, a general nonwoven fabric, polyurethane, a porous fluororesin, and the like can be used without particular limitation. The polishing pad is preferably grooved such that a polishing liquid is accumulated.

[0080] Regarding the polishing conditions, for example, the rotation speed of the polishing table (platen) and the carrier (head) is preferably 10 rpm (0.17 s.sup.-1) or more and 500 rpm (8.33 s.sup.-1) or less. The pressure (polishing pressure) applied to the substrate having the object to be polished is preferably 0.5 psi (3.45 kPa) or more and 10 psi (68.9 kPa) or less.

[0081] The method for supplying the polishing composition to the polishing pad is not particularly limited, and for example, a method for continuously supplying the polishing composition by a pump or the like is adopted. This supply amount is not limited, but it is preferable that the surface of the polishing pad is covered with the polishing composition according to the present disclosure.

[0082] The polishing composition according to the present disclosure may be a single-component

type or a multi-component type including a two-component type. In addition, the polishing composition according to the present disclosure may be prepared by diluting a stock solution of the polishing composition with a diluent such as water to, for example, 2 times or more and 100 times or less, usually 3 times or more and 50 times or less on a volume basis.

[Number of Scratches]

[0083] As described above, the polishing composition according to the present disclosure can reduce scratches on the surface of the polished object to be polished.

[0084] In the present disclosure, the number of scratches on the surface of the polished object to be polished is preferably as small as possible. Specifically, the number of scratches is practically 25 or less, preferably 20 or less, more preferably 15 or less, still more preferably 10 or less, and particularly preferably less than 10. In the present specification, as the number of scratches, a value measured by the method described in Examples is adopted.

[0085] Although the embodiments of the present disclosure have been described in detail, this is illustrative and exemplary and not restrictive, and it is clear that the scope of the present disclosure should be interpreted by the appended claims.

[0086] The present disclosure includes the following forms and aspects: [0087] 1. A polishing composition comprising abrasive grains in which an average primary particle size is 1 nm or more and 150 nm or less, and a dispersing medium, in which an aggregation rate of the abrasive grains is 40% or more and less than 75%; [0088] 2. The polishing composition according to the above 1., in which the aggregation rate of the abrasive grains is 40% or more and 55% or less; [0089] 3. The polishing composition according to the above 1, or 2., further comprising an inorganic salt; [0090] 4. The polishing composition according to the above 3., in which the inorganic salt is at least one selected from the group consisting of ammonium sulfate, ammonium nitrate, ammonium carbonate, potassium sulfate, potassium nitrate, and potassium carbonate; [0091] 5. The polishing composition according to any one of the above 1. to 4., further comprising an organic onium salt; [0092] 6. The polishing composition according to the above 5., in which the organic onium salt is at least one of a tetraalkylammonium salt represented by Chemical Formula 1 below or a tetraalkylphosphonium salt represented by Chemical Formula 2 below,

[Chem. 2]

[NR.sup.1R.sup.2R.sup.3R.sup.4].sup.+A.sup.- Chemical Formula 1

[PR.sup.5R.sup.6R.sup.7R.sup.8].sup.+X.sup.- Chemical Formula 2 [0093] in the Chemical Formula 1 and the Chemical Formula 2, [0094] R.sup.1 to R.sup.8 each independently represent an unsubstituted alkyl group having 1 or more and 4 or less carbon atoms, and [0095] A.sup.- and X.sup.- each independently represent a monovalent anion; [0096] 7. The polishing composition according to the above 6., in which the organic onium salt is a tetraalkylammonium salt represented by the Chemical Formula 1; [0097] 8. The polishing composition according to the above 6, or 7., in which A.sup.- and X.sup.- in the Chemical Formula 1 and the Chemical Formula 2 represent hydroxide ions; [0098] 9. The polishing composition according to any one of the above 6. to 8., in which the tetraalkylammonium salt represented by the Chemical Formula 1 is at least one selected from the group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetra n-butylammonium hydroxide; [0099] 10. The polishing composition according to any one of the above 1. to 9., in which a zeta potential of the abrasive grains in the polishing composition is negative; [0100] 11. The polishing composition according to the above 10., in which the abrasive grains are anionically modified colloidal silica; [0101] 12. The polishing composition according to any one of the above 1. to 11., in which pH is less than 7.0; [0102] 13. The polishing composition according to any one of the above 1. to 12., which is used for polishing an object to be polished containing silicon oxide; [0103] 14. A polishing method comprising: polishing an object to be

polished containing silicon oxide by using the polishing composition according to any one of the above 1. to 13.; and [0104] 15. A method for manufacturing a semiconductor substrate, the method comprising: polishing a semiconductor substrate containing silicon oxide by the polishing method according to the above 14.

EXAMPLES

[0105] The present disclosure will be described in more detail with reference to the following Examples and Comparative Examples. However, the technical scope of the present disclosure is not limited only to the following examples. Unless otherwise specified, “%” and “part” mean “% by mass” and “part by mass”, respectively. In addition, in the following Examples, unless otherwise specified, operations were performed under conditions of room temperature (20° C. or higher and 25° C. or lower)/relative humidity of 40% RH or higher and 50% RH or lower. Each physical property was measured as follows.

<Average Primary Particle Size and Average Secondary Particle Size of Abrasive Grains>

[0106] The average primary particle size of the abrasive grains was calculated from the specific surface area of the abrasive grains measured by the BET method using “Flow Sorb II 2300” manufactured by Micromeritics and the density of the abrasive grains. The average secondary particle size of the abrasive grains was measured as the volume average particle size (volume-based arithmetic average diameter; M_v) by a dynamic light scattering particle size/particle size distribution apparatus UPA-UT151 (manufactured by Nikkiso Co., Ltd.).

<Zeta Potential of Abrasive Grains>

[0107] The zeta potential of the abrasive grains in the polishing composition was measured using a zeta potential measuring apparatus (instrument name “ELS-Z2”) manufactured by Otsuka Electronics Co., Ltd.

<pH of Polishing Composition>

[0108] The pH of the polishing composition was measured by a pH meter (manufactured by HORIBA, Ltd., model number: LAQUA).

<Electrical Conductivity of Polishing Composition>

[0109] The electrical conductivity (EC) of the polishing composition was measured by a tabletop electrical conductivity meter (manufactured by HORIBA, Ltd., model number: DS-71 LAQUA (registered trademark)).

<Manufacturing of Sulfonic Acid-Immobilized Colloidal Silica>

[0110] According to the following procedure, sulfonic acid-modified colloidal silica as abrasive grains was obtained.

(Step of Preparing Raw Material Colloidal Silica Dispersion (Non-Modified Silica Particles))

[0111] In a flask, 4080 g of methanol, 610 g of water, and 168 g of a 29% by mass aqueous ammonia solution were mixed, the liquid temperature was maintained at 20° C., and a mixed liquid of 135 g of methanol and 508 g of tetramethoxysilane (TMOS) was added dropwise thereto for a dropwise addition time of 25 minutes. Thereafter, hot concentrated water substitution was performed under the condition of pH7 or more to obtain 1000 g of a 19.5% by mass silica sol (average primary particle size: 34 nm).

(Surface Modification Step)

[0112] Subsequently, to 1000 g of the silica sol obtained above (195 g in terms of silica solid content), 1.2 g of 3-mercaptopropyltrimethoxysilane (MPS, silane coupling agent, product name: KBM-803, manufactured by Shin-Etsu Chemical Co., Ltd.) (silane coupling agent concentration based on the total mass of silica solid content: 0.6% by mass) separately mixed with 4.8 g of methanol was added dropwise at a flow rate of 1 mL/min. Thereafter, heating was performed, and pure water replacement was performed for 3 hours after boiling.

[0113] Next, for cooling, the reaction solution was allowed to stand overnight, 0.0343 g of 30% by mass hydrogen peroxide water (3 mol with respect to 1 mol of the silane coupling agent) was added thereto, and the mixture was boiled again. Thereafter, pure water replacement was performed for 2

hours, followed by cooling to room temperature (25° C.) to obtain sulfonic acid-modified colloidal silica.

Example 1

<Preparation of Polishing Composition>

[0114] Sulfonic acid-modified colloidal silica (surface-modified silica 1, average primary particle size: 34 nm) as abrasive grains was added to water as a dispersing medium to have a final concentration of 2% by mass. Furthermore, ammonium sulfate (manufactured by Tokyo Chemical Industry Co., Ltd.) as an inorganic salt was added such that the final concentration becomes 6,000 ppm by mass, tetraethylammonium hydroxide (TEAH, manufactured by Tokyo Chemical Industry Co., Ltd.) as an organic onium salt was added such that the final concentration becomes 150 ppm by mass, and stirring and mixing were performed (stirring temperature: 25° C., stirring time: 20 minutes). The pH of the polishing composition was adjusted to 2.0 using nitric acid to complete a polishing composition 1.

Example 2

[0115] A polishing composition 2 was prepared in the same manner as in Example 1 except that the addition amount of tetraethylammonium hydroxide was changed to 100 ppm by mass.

Example 3

[0116] A polishing composition 3 was prepared in the same manner as in Example 1 except that the addition amount of tetraethylammonium hydroxide was changed to 1,000 ppm by mass.

Comparative Example 1

[0117] A comparative polishing composition 1 was prepared in the same manner as in Example 1 except that ammonium sulfate and tetraethylammonium hydroxide were not used.

Comparative Example 2

[0118] A comparative polishing composition 2 was prepared in the same manner as in Example 1 except that tetraethylammonium hydroxide was not used.

Comparative Example 3

[0119] A comparative polishing composition 3 was prepared in the same manner as in Example 1 except that n-pentylamine was added to the polishing composition such that the final concentration becomes 2,000 ppm by mass instead of tetraethylammonium hydroxide.

[0120] The configurations of the polishing compositions of each example and each comparative example are shown in Table 1 below. In Table 1 below, “-” indicates that the corresponding agent is not used.

TABLE-US-00001		TABLE 1		Abrasive grains		Average		Inorganic salt		Organic onium salt			
Polishing	Concen-	primary	secondary	Concen-	Concen-	composition	Polishing	tration	particle				
particle	Zeta	tration	tration	pH	EC	composition	(% by	size	size	potential	(ppm	(ppm	adjusting
No.	mass)	(nm)	(nm)	(mV)	Type	by mass)	Type	by mass)	agent	pH	cm)	Example	1 1 2 34 70 -25
Ammonium	6000	TEAH	150	Nitric	2.0	14	sulfate	acid	Example	2 2 2 34 70 -30	Ammonium	6000	
TEAH	100	Nitric	2.0	14	sulfate	acid	Example	3 3 2 34 70 -21	Ammonium	6000	TEAH	1000	
Nitric	2.0	14	sulfate	acid	Comparative	Comparative	1 2 34 70 -40	—	—	—	—	Nitric	2.0 6
Example	1	acid	Comparative	Comparative	2 2 34 70 -35	Ammonium	6000	—	—	Nitric	2.0 14		
Example	2	sulfate	acid	Comparative	Comparative	3 2 34 70 -28	Ammonium	6000	n-	2000	Nitric		
2.0	14	Example	3	sulfate	pentylamine	acid							

[Evaluation]

<Aggregation Rate of Abrasive Grains>

[0121] The aggregation rate of the abrasive grains was measured by the following method: [0122]

1) 30.0 g of the polishing composition as a sample was weighed into a polypropylene container (centrifuge tubes) manufactured by Beckman Coulter, Inc., and the upper lid was firmly closed;

[0123] 2) Using a “Centrifuge Avanti HP-30I” manufactured by Beckman Coulter, Inc., continuous operation was performed at a temperature of 25° C. and 15,000 rpm for 15 minutes; [0124] 3) By centrifugation, the moisture at the top of the container and the abrasive grains at the bottom were

separated, and the abrasive grains were compressed and settled at the bottom of the container (referred to as a “cake”); [0125] 4) This precipitated cake was dried at 200° C. for 24 hours using “ELECTRIC MUFFLE FURNACES KM-420” manufactured by Advantec Co., Ltd., and then the weight of the obtained solid was measured; and [0126] 5) Separately, using an aqueous dispersion of abrasive grains, the weight of the dried solid measured by the procedures of 1) to 4) above was defined as (B), the weight of the dried solid measured using the polishing composition as a sample was defined as (A), and the aggregation rate was calculated by the following Formula.

[00002] $\text{Aggregationrate(\%)} = [(A) / (B)] \times 100$ [MathematicalFormula2]

[0127] In addition, (A) and (B) may contain a trace amount of additive, but a value calculated by Formula (1) without considering this was taken as the aggregation rate of the abrasive grains.

<Polishing Removal Rate>

[0128] A silicon wafer (200 mm, blanket wafer) having a TEOS type silicon oxide (SiO₂) film having a thickness of 10,000 Å formed on a surface thereof was prepared as an object to be polished, and polishing was performed under the following conditions.

(Polishing Apparatus and Polishing Conditions)

[0129] Polishing apparatus: CMP single-side polishing apparatus for 200 mm, manufactured by Applied Materials Inc., Mirra [0130] Polishing pad: polyurethane pad IC1010 manufactured by NITTA DuPont, Inc. [0131] Polishing pressure: 2.5 psi (1 psi=6894.76 Pa) [0132] Rotation speed of polishing table: 47 rpm [0133] Rotation speed of head (carrier): 43 rpm [0134] Supply of polishing composition: continuous flow [0135] Polishing composition supply amount: 200 mL/min [0136] Polishing time: 60 seconds

[0137] For the object to be polished, the thickness before and after polishing was determined with an optical film thickness measuring instrument (ASET-f5x: manufactured by KLA-Tencor Corporation). The film thickness was determined by an optical film thickness measuring instrument (ASET-f5x: manufactured by KLA-Tencor Corporation).

[0138] In the object to be polished, the polishing removal rate in each object to be polished was calculated by dividing the difference in film thickness before and after polishing [(thickness before polishing)–(thickness after polishing)] by the polishing time.

<Number of Scratches>

[0139] The number of scratches on the surface of the silicon oxide film after polishing was measured by measuring coordinates of the entire surface of both surfaces of the object to be polished (excluding the outer circumference of 2 mm) using a wafer inspection apparatus “Surfscan (registered trademark) SP2” manufactured by KLA-Tencor Corporation and observing all the measured coordinates with Review-SEM (RS-6000, manufactured by Hitachi High-Technologies Corporation). Incidentally, scratches on the substrate surface having a depth of 10 nm or more and less than 100 nm, a width of 100 nm or more and less than 500 nm, and a length of 80 nm or more were counted as scratches. The smaller the number of scratches, the more preferable. The number of scratches is practically 25 or less, preferably 20 or less, more preferably 15 or less, still more preferably 10 or less, and particularly preferably less than 10.

[0140] The evaluation results of the abrasive grain aggregation rate, the polishing removal rate, and the number of scratches are shown in the following Table 2.

TABLE-US-00002 TABLE 2 Abrasive grain Polishing Number of aggregation removal rate scratches rate (%) (Å/min) (pieces) Example 1 53 322 8 Example 2 42 325 8 Example 3 70 316 15 Comparative 100 150 60 Example 1 Comparative 81 330 36 Example 2 Comparative 37 245 7 Example 3

[0141] As is apparent from Table 2 above, it was found that in a case where the polishing composition of Example was used, the polishing removal rate of the silicon oxide film was high, and scratches on the surface of the polished silicon oxide film could be reduced. It was found that in a case where the polishing compositions of Comparative Examples 1 and 2 in which the

aggregation rate of the abrasive grains is high were used, scratches on the surface of the silicon oxide film increased. In addition, it was found that in a case where the polishing composition of Comparative Example 3 in which the aggregation rate of the abrasive grains is low was used, the polishing removal rate of the silicon oxide film was reduced.

[0142] The present application is based on Japanese Patent Application No. 2024-036950 filed on Mar. 11, 2024, the disclosure content of which is incorporated herein by reference in its entirety.

Claims

1. A polishing composition comprising: abrasive grains in which an average primary particle size is 1 nm or more and 150 nm or less; and a dispersing medium, wherein an aggregation rate of the abrasive grains is 40% or more and less than 75%.
 2. The polishing composition according to claim 1, wherein the aggregation rate of the abrasive grains is 40% or more and 55% or less.
 3. The polishing composition according to claim 1, further comprising an inorganic salt.
 4. The polishing composition according to claim 3, wherein the inorganic salt is at least one selected from a group consisting of ammonium sulfate, ammonium nitrate, ammonium carbonate, potassium sulfate, potassium nitrate, and potassium carbonate.
 5. The polishing composition according to claim 1, further comprising an organic onium salt.
 6. The polishing composition according to claim 5, wherein the organic onium salt is at least one of a tetraalkylammonium salt represented by Chemical Formula 1 below or a tetraalkylphosphonium salt represented by Chemical Formula 2 below:
[Chem. 1]
[NR.sup.1R.sup.2R.sup.3R.sup.4].sup.+A.sup.- Chemical Formula 1
[PR.sup.5R.sup.6R.sup.7R.sup.8].sup.+X.sup.- Chemical Formula 2 in the Chemical Formula 1 and the Chemical Formula 2, R.sup.1 to R.sup.8 each independently represent an unsubstituted alkyl group having 1 or more and 4 or less carbon atoms, and A.sup.- and X.sup.- each independently represent a monovalent anion.
 7. The polishing composition according to claim 6, wherein the organic onium salt is a tetraalkylammonium salt represented by the Chemical Formula 1.
 8. The polishing composition according to claim 6, wherein A.sup.- and X.sup.- in the Chemical Formula 1 and the Chemical Formula 2 represent hydroxide ions.
 9. The polishing composition according to claim 6, wherein the tetraalkylammonium salt represented by the Chemical Formula 1 is at least one selected from the group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetra n-butylammonium hydroxide.
 10. The polishing composition according to claim 1, wherein a zeta potential of the abrasive grains in the polishing composition is negative.
 11. The polishing composition according to claim 10, wherein the abrasive grains are anionically modified colloidal silica.
 12. The polishing composition according to claim 1, wherein pH is less than 7.0.
 13. The polishing composition according to claim 1, which is used for polishing an object to be polished containing silicon oxide.
 14. A polishing method comprising: polishing an object to be polished containing silicon oxide by using the polishing composition according to claim 1.
 15. A method for manufacturing a semiconductor substrate, the method comprising: polishing a semiconductor substrate containing silicon oxide by the polishing method according to claim 14.
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